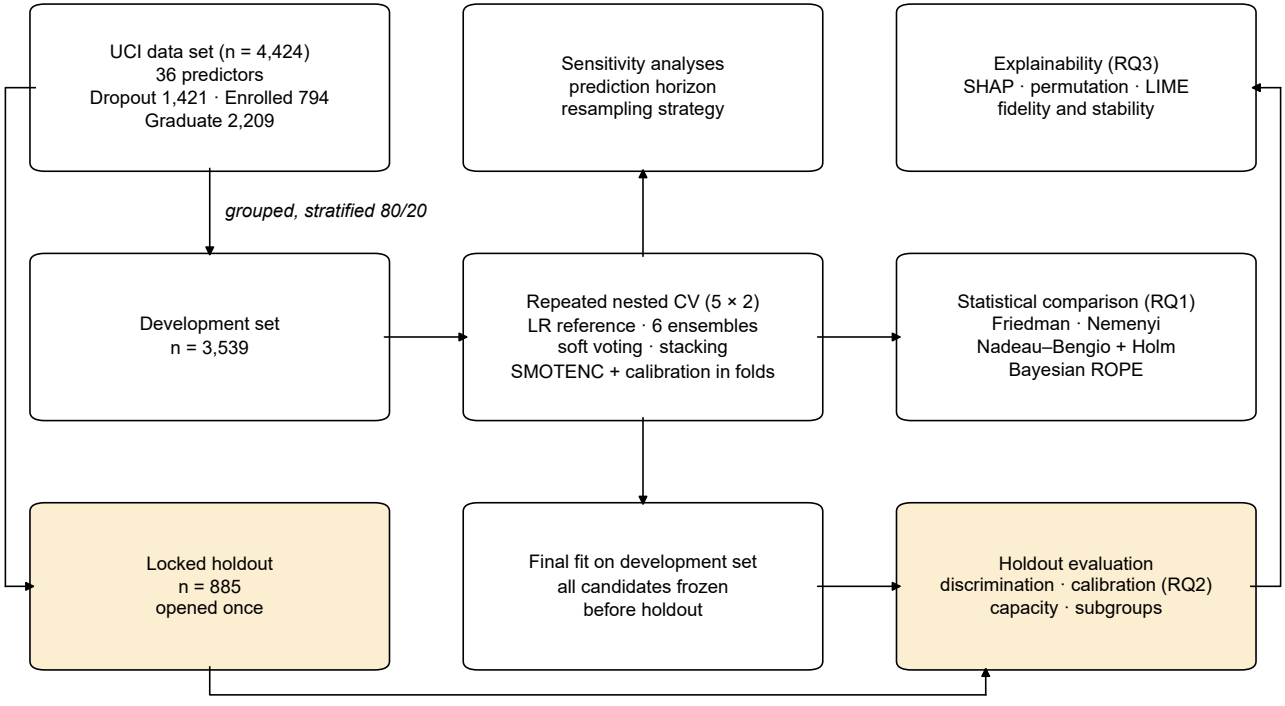
UCI data set (n = 4,424)
36 predictors
Dropout 1,421 · Enrolled 794
Graduate 2,209
grouped, stratified 80/20
Development set
n = 3,539
Locked holdout
n = 885
opened once
Repeated nested CV (5 × 2)
LR reference · 6 ensembles
soft voting · stacking
SMOTENC + calibration in folds
Final fit on development set
all candidates frozen
before holdout
Sensitivity analyses
prediction horizon
resampling strategy
Statistical comparison (RQ1)
Friedman · Nemenyi
Nadeau–Bengio + Holm
Bayesian ROPE
Holdout evaluation
discrimination · calibration (RQ2)
capacity · subgroups
Explainability (RQ3)
SHAP · permutation · LIME
fidelity and stability